

Longitudinal Tissue Phantoms for Point-of-Care Settings

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Abstract

Work in biomedical research has long relied on assumptions we test directly here. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. The measurements reported here were collected over twelve independent trials under controlled conditions. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

1. Introduction

Residual error grows approximately linearly with the number of coupled parameters. Sensitivity to the smoothing window proved modest across the range we examined. We release the processing scripts and intermediate artefacts to support independent verification. Taken together, these findings argue for treating the two regimes separately in future analyses.

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Taken together, these findings argue for treating the two regimes separately in future analyses. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Sensitivity to the smoothing window proved modest across the range we examined. Longer observation windows narrow the confidence interval but do not shift the central estimate. Residual error grows approximately linearly with the number of coupled parameters. The reduced model retains most of the explanatory power at a fraction of the computational cost. Neither of the alternative formulations improved on the simple additive model in our setting. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

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2. Related Work

Calibration drift over the trial period remained below the threshold set out in the protocol. Residual error grows approximately linearly with the number of coupled parameters. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Longer observation windows narrow the confidence interval but do not shift the central estimate. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Taken together, these findings argue for treating the two regimes separately in future analyses.

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3. Materials and Methods

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4. Experimental Setup

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